

B. Tech. (3rd Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Big Data and Analytics, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BCSE-504L (Object Oriented Programming using JAVA Lab)

L	T	P	Continuous evaluation	60
0	0	2	End-of-semester exam	40
			Total marks	100

Course Objectives

1. To introduce the fundamentals of Object-Oriented Programming and Java programming environment. To acquire knowledge about Java programming language.
2. To develop programming skills using classes, objects, inheritance, polymorphism, packages, and interfaces.
3. To familiarize students with exception handling and multithreading concepts for developing robust applications.
4. To enable students to design and develop GUI-based applications using AWT, and event handling techniques.

List of Experiments

1. Consider the user has N eggs. Then display the no of eggs in gross (144 eggs make one gross), the number of eggs in a dozen (12 eggs make one dozen), and the no of eggs that are left out remaining. The total no of eggs can be obtained as input through the console. The program should display how many gross, how many dozen, and how many leftover eggs the user has. For example, if the input is 1342 eggs, the program should respond with: "Your number of eggs is 9 gross, 3 dozen, and 10."
2. Get two numbers as input from the user through the console, swap the values of the two numbers without using a temporary variable, and display the same.
3. Tax slabs for general 0 to 1,80,000 No tax, 1,80,001 to 5,00,000 10%, 5,00,001 to 8,00,000 20%, Above 8,00,000 30%. Income tax slabs for Women 0 to 1,90,000 No tax, 1,90,001 to 5,00,000 10%, 5,00,001 to 8,00,000 20%, Above 8,00,000 30%. Write if statements to achieve this. Make sure that you indent the code well so that it is readable.
4. Create a class called Calculator that has different add methods. Overload the methods with different parameters.
5. Demonstrate the concept of a user-defined package using the following scenario: Create a new project in which you create a package named org.animals. In that, create various classes such as Lion, Monkey, and Elephant. In each class, create data members such as colour, weight, and age. Create methods like is Vegetarian, canClimb, getSound. Create another project, and in it create a package called zoo. Create a class called VandalurZoo, create objects for the animals in the zoo, and print the characteristics of each animal.
6. WAP to demonstrate the concept of Method Overriding.
7. WAP to implement the concept of Interface and Abstract class.
8. WAP to implement the concept of Exception Handling by using try, catch, and finally.
9. WAP to create three threads and a for loop from 1 to 10 and utilise context switching, sleep and other important methods under the Thread class.
10. WAP to demonstrate the concept of GUI with Event handling using Swing and AWT.

Projects :

- Learning Management System (LMS)
- Online Shopping System (E-Commerce Application)

Course Outcomes: After completion of this course, students will be able to:

1. Apply Java programming fundamentals, operators, control structures, and input/output techniques to solve computational problems.
2. Design and develop object-oriented applications using classes, objects, methods, constructors, overloading, and packages.
3. Implement object-oriented principles such as inheritance, polymorphism, abstract classes, and interfaces in software development.
4. Develop robust Java applications using exception handling and multithreading concepts.
5. Design and implement event-driven graphical user interfaces using AWT and Swing for interactive applications.

Text Books:

1. “Core Java Volume-I and II” 2nd edition-Sun Micro System.
2. Java: The Complete Reference, Herbert Schildt, McGraw Hill Education.

Reference Books:

1. Programming with Java, E. Balaguruswamy, McGraw Hill Education.
2. Rick Dranell , “HTML 4 Unleashed”, Second edition, Tech media publication.
3. Head First Java, O'Reilly Media.
4. Object-Oriented Programming with Java, McGraw Hill Education.
5. Java How to Program, Pearson Education.